

Quantitative MicroCT Analysis of Taurine and Leptin Effects on Bone Remodeling

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Introduction: Bone tissue is adaptable, and its structure responds to external stimuli such as mechanical loading. However, with aging, this responsiveness declines, resulting in more frail bones and consequently, a higher degree of fractures. Resultingly, the broader impact is that patients must deal with more pain and surgeries as they age. Therefore, identifying molecular treatments that could preserve or restore bone adaptability is therefore an important goal in preventing related bone deterioration.

Taurine:

Taurine supplementation has been linked with stronger bones and a decreased bone deterioration as mice age, increasing life span altogether [1]. In particular, after performance of a three-point bending test, femur maximal load improved in taurine treated mice compared to control mice. Due to the ability of Taurine to help maintain bone structure, our lab sought out to explore how it affects the response of bones to mechanical loading.

Leptin:

As a response to metabolic needs such as starvation or aging, the body generally relies on regulated bone marrow adipose tissue (rBMAT), located in the mid to proximal tibia, as a means to support bone health and maintain energy balance. In contrast, constitutive bone marrow adipose tissue (cBMAT), located in the distal tibia, generally maintains a reserve of itself and is stable in its quantity, acting as a ‘padding’ to support bone structure. Once depleted, however, cBMAT could be a great target for net treatment / leveraging this reserve for energy.

Leptin is a hormone produced by adipose cells to regulate satiety, and supplementation of such was identified to function in depleting cBMAT [2]. Therefore, our lab sought out to use leptin as a means to deplete all cBMAT, and in turn determine whether cBMAT can be used to aid in local bone formation.

When performing these experiments, our lab chose to use mice as the experimental model because their bone remodeling process closely parallels those of humans, and they allow for

precise control over systemic treatments and mechanical loading conditions. As opposed to cell culture systems, mice provide an intact physiological environment where hormonal, metabolic, and mechanical factors interact to regulate bone adaptation. Additionally, the ethics of fat reduction using Taurine combined with the limitations in performing controlled mechanical loading experiments makes mice the best fit for our projects.

To create bone scans, our lab utilized micro computed tomography (microCT) imaging to quantify structural outcomes and create comparisons among experimental groups. MicroCT provides high-resolution, three-dimensional visualization of bone architecture, allowing for precise quantification of cortical and trabecular geometry.

Throughout this semester, my primary responsibility that overlapped between both the Taurine and Leptin projects involved microCT-based data analysis of these scans, as the derived bone structure parameters constituted the outcome of these studies.

Methods: For analyzing bones in both projects, I utilized a combination of the BoneJ plugin of Fiji software and ScanCo Evaluation software.

BoneJ:

BoneJ is an open-source collection of plugins for Fiji/ImageJ that provides various tools for measuring trabecular bone and whole bone shape from CT scans, which allows quantification of skeletal health and structure without the need of a microCT. BoneJ analysis could be done from my laptop, and made the process a lot more efficient with a more user-friendly interface. I used BoneJ to calculate cortical bone parameters along the bone length, due to its ease of use, to realize potential changes along a larger region as opposed to one location with the ScanCo procedure. Additionally, since not every scan started on the same slice, I normalized the tibia length so all samples could be comparable.

ScanCo:

ScanCo Evaluation is a commercial software that performs high-resolution 3D analysis, allowing for precise measurements of physical parameters. It is generally more comparable among scientific papers and studies, serving as the gold standard for analysis of the trabecular and cortical bone parameters. However, ScanCo Evaluation requires a workstation readily available, the procedure is more complex, and generally analyzes smaller regions. ScanCo was used more for the taurine project, as I was only analyzing 100 slices of the midshaft and wanted exactness in the generation of physical parameters.

The use of both BoneJ and ScanCo software allowed for complementary analysis strategies within this study. BoneJ enabled efficient assessment of cortical parameters along the full analyzed length of the tibia, providing spatial resolution that would be difficult to achieve using ScanCo alone. In contrast, ScanCo was used to generate high-precision measurements within defined regions of interest, particularly for trabecular analysis. Together, these approaches ensured both spatial coverage and measurement accuracy across projects.

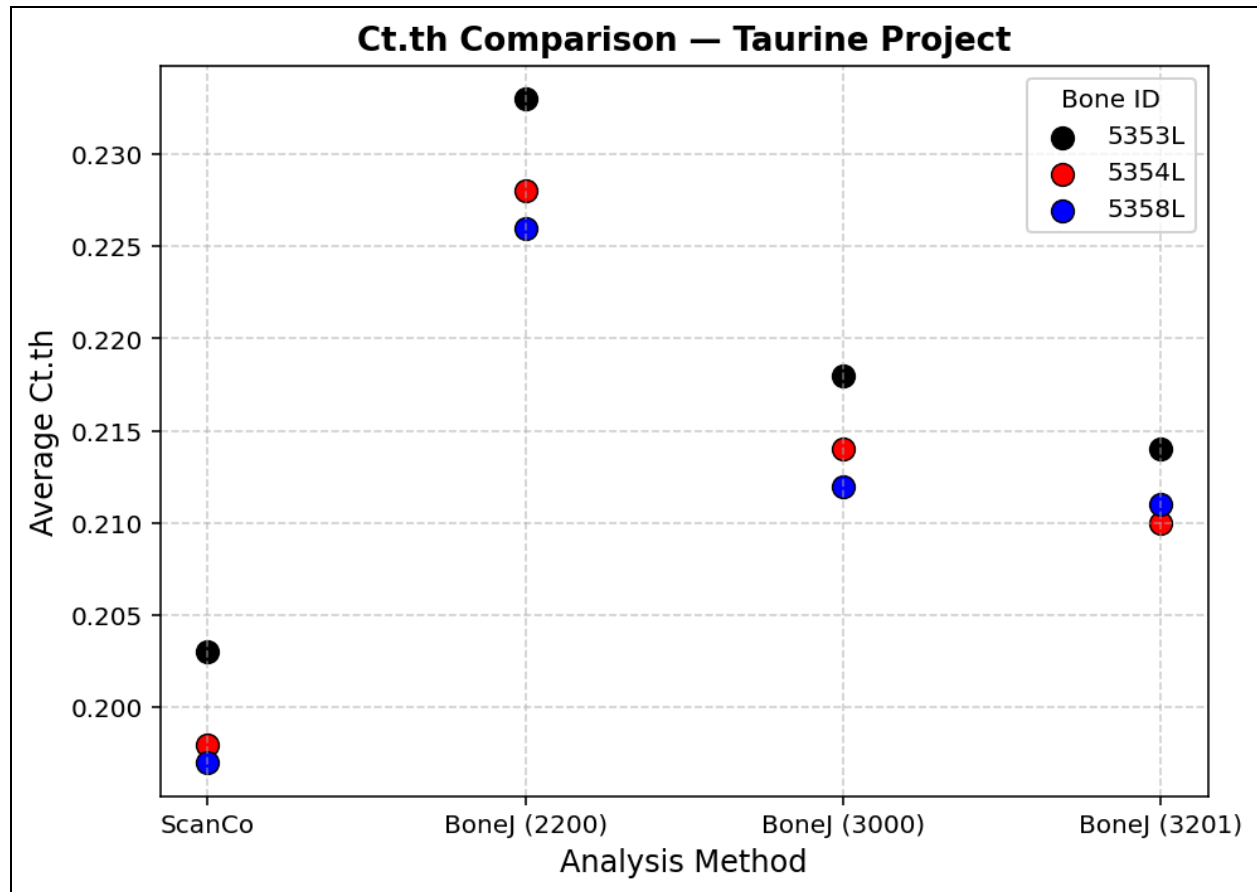


Figure 1: Comparison of Ct.Th values for using ScanCo vs. BoneJ at various thresholds for 3 different bones

Since BoneJ is generally less of a benchmark software for analysis, I performed a validation test following the trabecular bone procedure to evaluate differences in cortical thickness values obtained using each software across varying BoneJ threshold settings. Specifically, cortical thickness values derived from BoneJ were compared against ScanCo outputs while systematically adjusting BoneJ threshold parameters to assess sensitivity to segmentation choice. This comparison allowed for evaluation of whether variations in threshold selection produced substantial deviations in measured cortical thickness.

The difference in average Ct.Th values held relatively consistent when comparing bones among each ScanCo and varying BoneJ threshold group. Across the tested threshold ranges, relative differences between samples were preserved, indicating that cortical thickness trends remained stable despite changes in threshold selection. These results suggest that BoneJ-derived cortical thickness measurements were comparable to ScanCo outputs for the purposes of relative group comparison within this study.

Regions Analyzed:

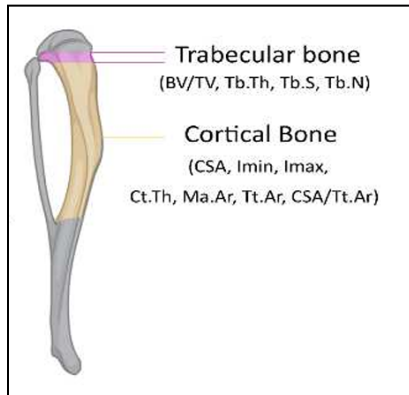


Figure 2: Regions of the tibia analyzed using microCT. The cortical midshaft (yellow) was analyzed with BoneJ, and the trabecular region (pink) was analyzed with ScanCo. Adapted from Ref. [3].

Figure 1 illustrates the regions of the tibia analyzed in this study.

The yellow region corresponds to the cortical midshaft, where structural parameters were quantified primarily using the BoneJ plugin in Fiji. The procedure follows identifying the Tibia-Fibula Junction (TFJ) as the starting slice, segmenting out the fibula, and analyzing from the start to the end slice.

The pink region represents the trabecular bone region analyzed using solely ScanCo software. These regions were selected to capture both cortical and trabecular adaptations in response to treatment and mechanical loading. The procedure follows identifying the first metaphyseal slice distal to the growth plate, moving 10 slices further, and analyzing the next 100 consecutive slices.

Upon completion of analyzing each region, I used python to create plots that created comparisons of the parameters we sought to analyze.

Parameters:

Cortical Thickness:

Cortical thickness (Ct.Th) represents the average thickness of the cortical bone shell at a given cross-section along the tibia. Because the cortical compartment bears the majority of mechanical load in long bones, Ct.Th is a primary determinant of bending and compressive strength. Increases in cortical thickness generally reflect enhanced bone formation or reduced bone resorption, resulting in a greater capacity to withstand applied mechanical forces. Mechanical loading is known to stimulate cortical thickening as an adaptive response, making Ct.Th a sensitive metric for evaluating mechanoadaptation. In the context of this study, changes in Ct.Th provide direct insight into how mechanical loading and molecular treatments influence the structural integrity of load-bearing bone.

Imax:

The maximum second moment of area (Imax) quantifies the resistance of a bone cross-section to bending about its strongest principal axis. This parameter reflects how bone material is distributed relative to the axis that experiences the greatest resistance to bending, making it a key indicator of structural strength in long bones such as the tibia

Because mechanical loading of the tibia induces directional bending, Imax is particularly informative for assessing functional adaptation to loading and treatment. Changes in Imax therefore provide insight into whether structural remodeling enhances the bone's ability to resist physiologically relevant bending forces.

Imin:

The minimum second moment of area (Imin) represents the resistance of a bone cross-section to bending about its weakest principal axis. While typically smaller than Imax, Imin is important for understanding structural robustness and directional symmetry of the bone.

Changes in Imin indicate whether adaptation occurs uniformly across multiple bending directions or is primarily localized along a dominant loading axis. Together, Imax and Imin parameters provide a more complete assessment of how treatments and mechanical loading influence bone geometry and mechanical competence.

Bone Volume:

Bone volume (BV) represents the total volume of mineralized bone tissue within a defined region of interest. This parameter reflects overall bone mass and provides a global measure of skeletal accrual or loss in response to mechanical and molecular stimuli. While bone volume alone does not fully capture mechanical competence, increases in BV generally indicate enhanced bone formation or reduced resorption within the analyzed region. In loading studies, changes in BV help distinguish whether observed structural adaptations arise from increased bone mass or from geometric redistribution of existing bone. In the context of this study, evaluating BV alongside cortical thickness and bending resistance parameters allows for interpretation of whether treatments and mechanical loading promote overall bone accrual or primarily alter bone geometry.

Cross Sectional Area:

Cross-sectional area (CSA) represents the total area of bone tissue within a transverse cross-section of the tibia. This parameter provides a measure of local bone mass and is directly related to the bone's ability to resist axial compressive loads. In the context of mechanical loading studies, CSA helps distinguish whether structural adaptation occurs through overall increases in bone mass or through redistribution of existing bone material. When interpreted alongside cortical thickness and bending resistance parameters, CSA contributes to a more complete understanding of how mechanical and molecular stimuli influence bone structure and mechanical competence.

Perimeter:

Perimeter represents the total length of the outer cortical boundary measured in a transverse cross-section of the tibia. This parameter reflects the overall external geometry of the bone and is influenced primarily by periosteal expansion. Changes in perimeter provide insight into how bone adapts through alterations in shape rather than thickness alone. In the context of mechanical loading and molecular treatment studies, perimeter helps distinguish whether structural adaptation occurs via outward expansion of the cortical surface or through internal redistribution of bone tissue. When considered alongside cross-sectional area and cortical thickness, perimeter contributes to a more complete characterization of cortical bone geometry and structural adaptation.

Results and Discussion:

Taurine Project:

For the taurine project, our lab created four groups of mice based on treatment and mechanical load. The resulting groups created were: Taurine Control, Vehicle Control, Taurine Loaded, Vehicle Loaded.

Two important parameters we fixated on for the taurine project were cortical thickness (Ct.Th) and Bone Volume.

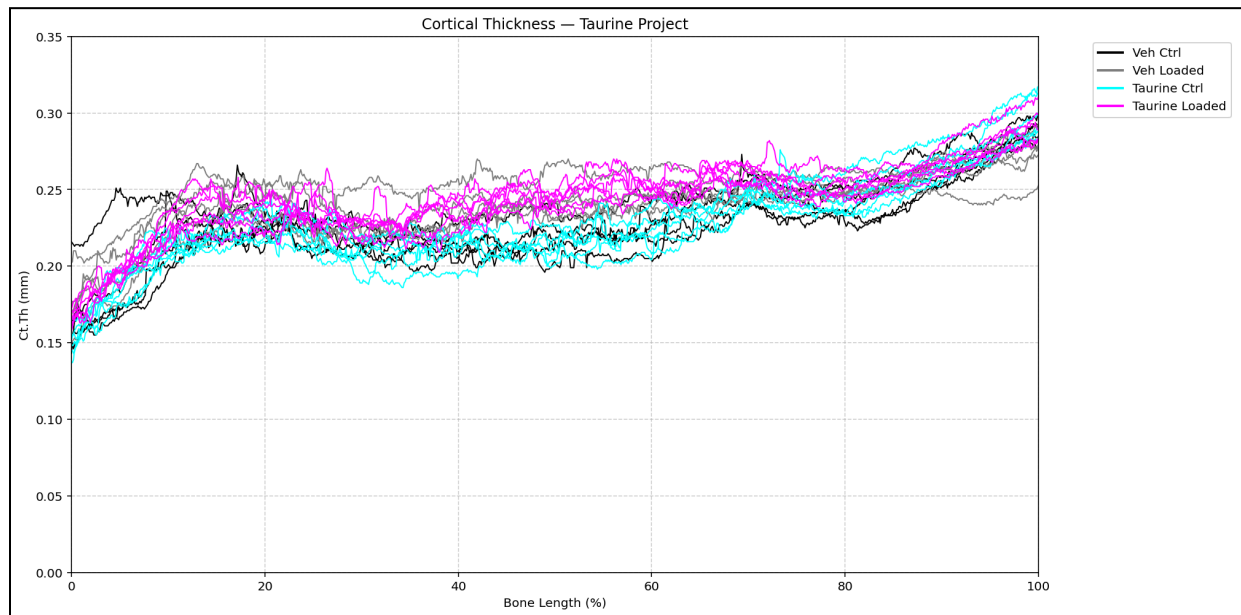


Figure 3: Cortical thickness for each bone across the bone length (normalized).

Cortical Thickness:

Figure 3 presents cortical thickness profiles along the normalized tibial length for taurine-treated and control groups under both loaded and non-loaded conditions. Across the analyzed region, bones subjected to mechanical loading exhibit higher cortical thickness values compared to non-loaded controls, regardless of taurine treatment.

In contrast, cortical thickness curves for taurine-treated groups largely overlap with their respective vehicle-treated counterparts within the same loading condition. This overlap is maintained across proximal, midshaft, and distal portions of the analyzed region, indicating that taurine treatment did not produce measurable changes in cortical thickness relative to control when mechanical loading conditions were matched.

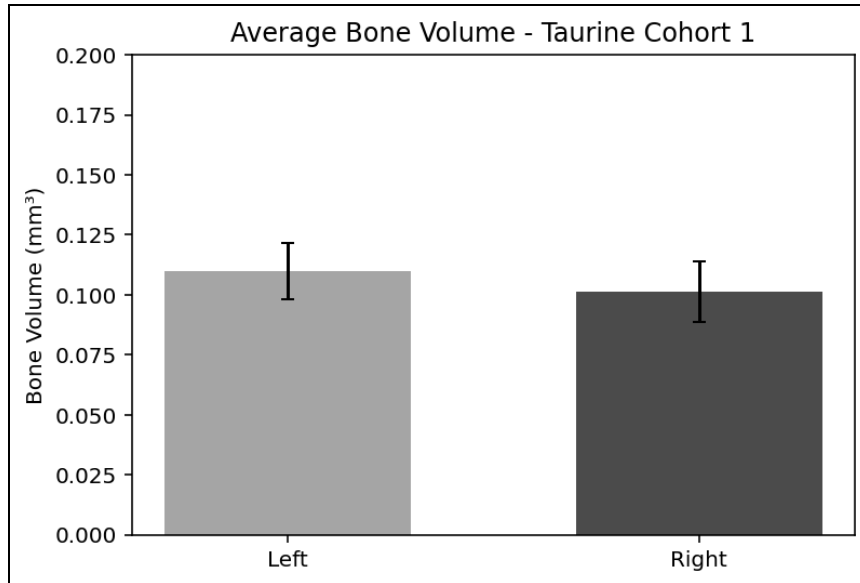


Figure 4: Average Bone volume for treated (left) vs. vehicle (right).

Bone Volume:

Figure 4 further compares cortical thickness profiles among the vehicle-loaded, vehicle-control, taurine-loaded, and taurine-control groups. The left group of bones are treated with taurine (including both loaded and control), while the right group of bones are vehicle (no treatment but include both loaded and control).

Across the length of the tibia, loaded groups consistently display greater cortical thickness than their non-loaded counterparts, independent of treatment. Cortical thickness values for taurine-loaded bones closely align with those of vehicle-loaded bones, while taurine-control bones similarly overlap with vehicle-control groups. No consistent separation between taurine and vehicle groups is observed along the normalized bone length, indicating that mechanical loading, rather than taurine supplementation, is the primary contributor to differences in cortical thickness.

Leptin Project:

For the leptin project, our lab created groups based on leptin or PBS treatment (control), Wild Type or Knockout, and 21 or 9 days of treatment for both male and female mice.

To clarify the distinction between Wild-Type (WT) and Knockout (KO) mice, KO mice with disrupted fat cell lipolysis were used to limit systemic energy availability. This model was used to assess whether leptin-related changes in bone structure depend on metabolic energy mobilization from adipose tissue.

The parameters we fixated on for the leptin project were cortical thickness (Ct.Th), cross sectional area (CSA), I_{max}, and I_{min}.

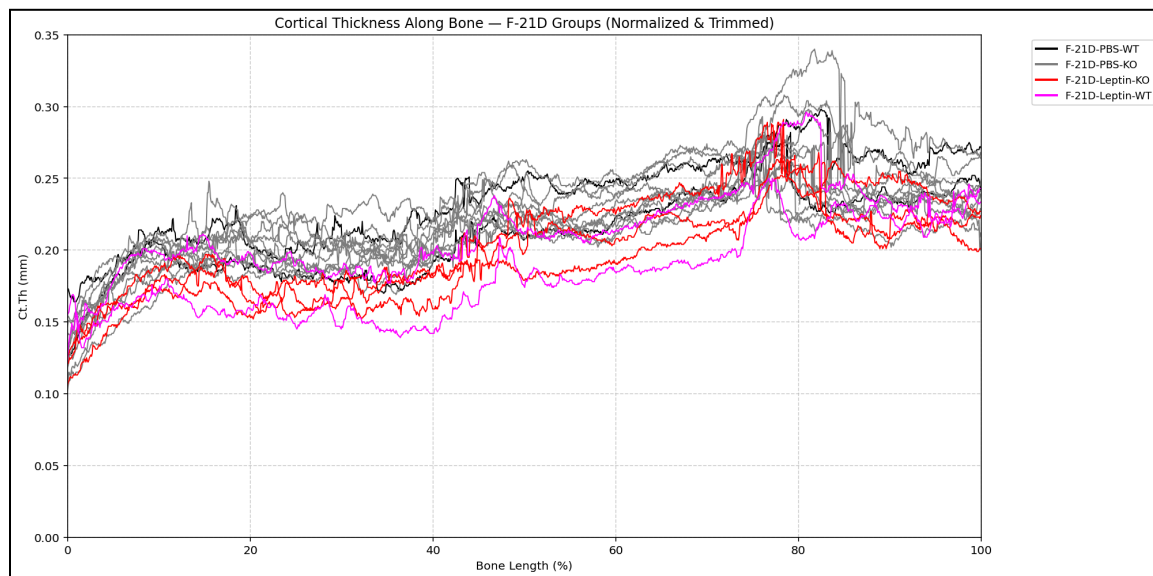


Figure 5: Cortical thickness for each 21 day treated female bone across the bone length (normalized).

Cortical Thickness:

From Figure 5, it is evident that 21-day leptin treatment resulted in a lower cortical thickness across the length of the tibia when compared to PBS-treated controls, regardless of knockout or wild-type group. This reduction is observed consistently along the normalized bone length, with leptin-treated curves residing below PBS-treated groups across most regions of the tibia. Additionally, the relative separation between leptin and PBS groups appears maintained throughout proximal, midshaft, and distal regions, indicating that the observed decrease in cortical thickness is not localized to a single anatomical region.

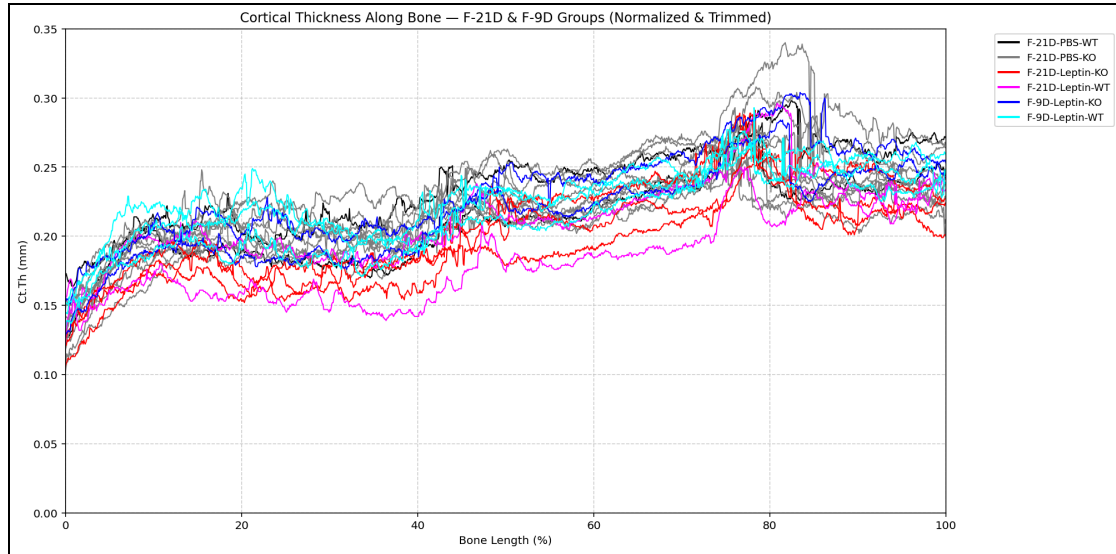


Figure 6: Cortical thickness for 21 day and 9D treated female bones across the bone length (normalized).

Cortical Thickness:

Figure 6 includes the same treatment groups as Figure 5, with the addition of 9-day leptin-treated bones. From this figure, the 9-day leptin groups consistently exhibit cortical thickness values that fall between those of the 21-day leptin-treated and PBS-treated groups across much of the bone length. This pattern is observed across proximal, midshaft, and distal regions, suggesting a graded relationship in cortical thickness corresponding to treatment duration. Notably, the 21-day leptin-treated groups display the lowest cortical thickness values, while PBS-treated groups maintain the highest values throughout most of the tibia.

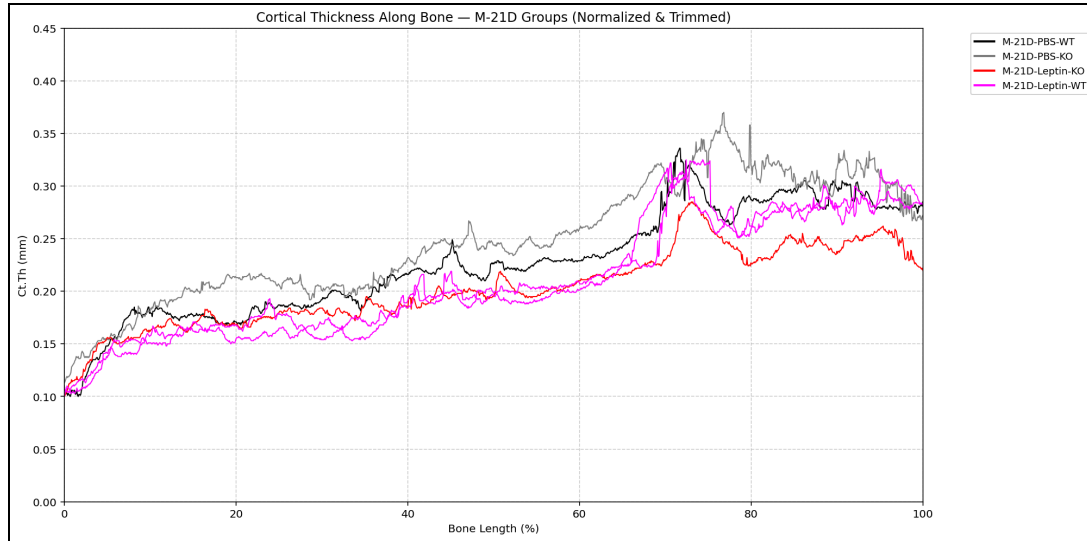


Figure 7: Cortical thickness for each 21 day treated male bone across the bone length (normalized).

Cortical Thickness:

Figure 7 represents the same analysis shown in Figure 5, but for male bones treated for 21 days. Similar to the female cohort, leptin-treated male bones exhibit lower cortical thickness values along the length of the tibia when compared to PBS-treated controls. This trend is consistent across the normalized bone length and is observed in both wild-type and knockout groups, indicating that the reduction in cortical thickness associated with leptin treatment is reproducible across sex.

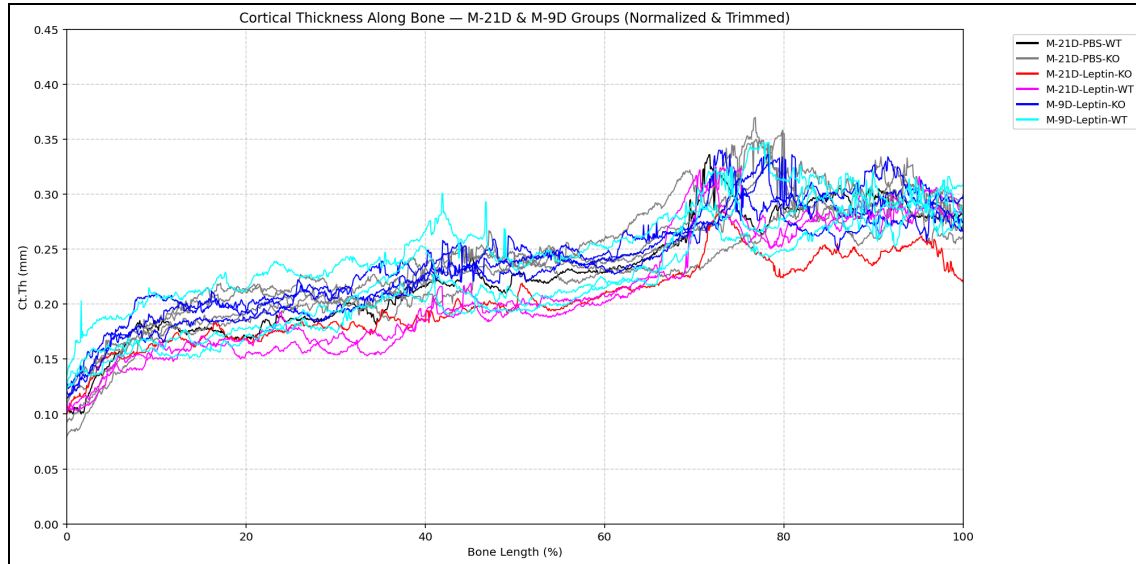


Figure 8: Cortical thickness for 21 day and 9D treated male bones across the bone length (normalized).

Cortical Thickness:

Figure 8 presents cortical thickness profiles for male bones treated with leptin for 9 and 21 days, alongside PBS controls. Consistent with the female cohort, the 9-day leptin-treated group exhibits cortical thickness values that are higher than those of the 21-day leptin-treated group but lower than PBS-treated controls across much of the tibial length. This ordering of groups is maintained throughout multiple regions of the bone, further supporting the observation that treatment duration influences the magnitude of cortical thickness reduction.

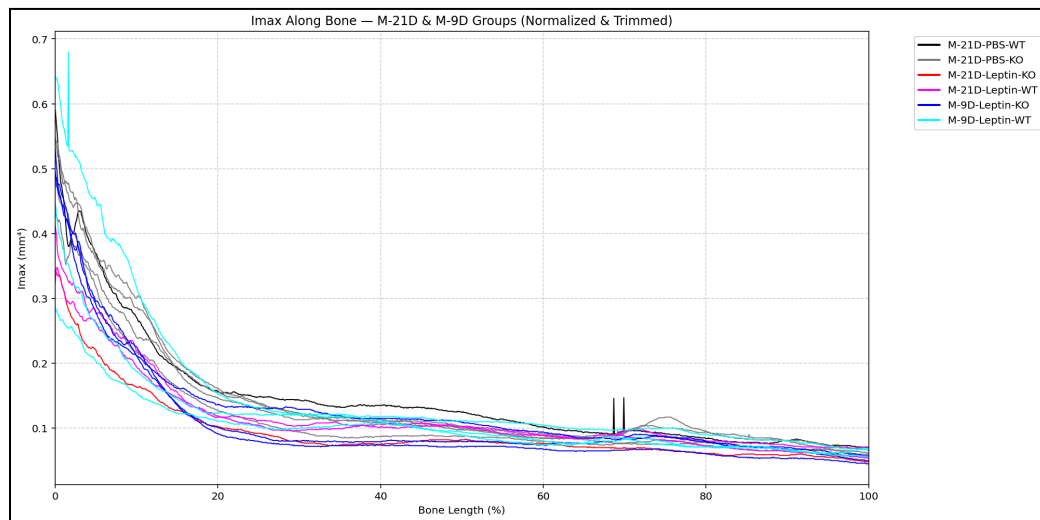


Figure 9: Imax for 21 day and 9 day treated male bones across the bone length (normalized).

I_{max}:

Figure 9 illustrates I_{max} values along the normalized bone length for both 21-day and 9-day treated male bones. Across the majority of the tibia, I_{max} profiles show substantial overlap among leptin-treated and PBS-treated groups, with no consistent separation observed between genotypes or treatment durations. While localized fluctuations are present, overall trends indicate that I_{max} remains relatively similar across groups throughout most regions of the bone.

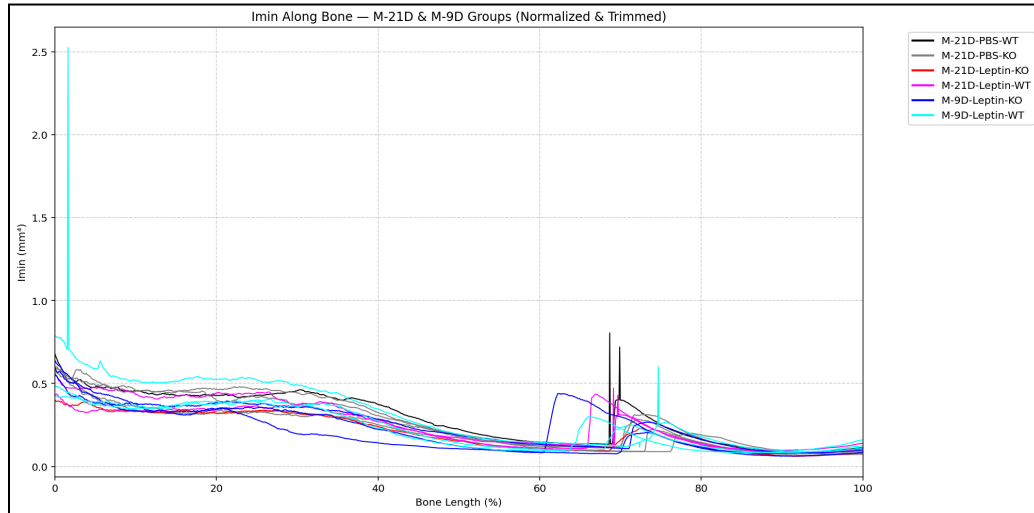


Figure 10: I_{min} for 21 day and 9 day treated male bones across the bone length (normalized).

I_{min}:

Similar to I_{max}, Figure 10 demonstrates that I_{min} values along the tibial length are largely overlapping among all treatment groups. Across proximal, midshaft, and distal regions, no consistent differences are observed between leptin-treated and PBS-treated bones or between wild-type and knockout groups. Although minor regional variability is present, the overall distribution of I_{min} values remains comparable across conditions.

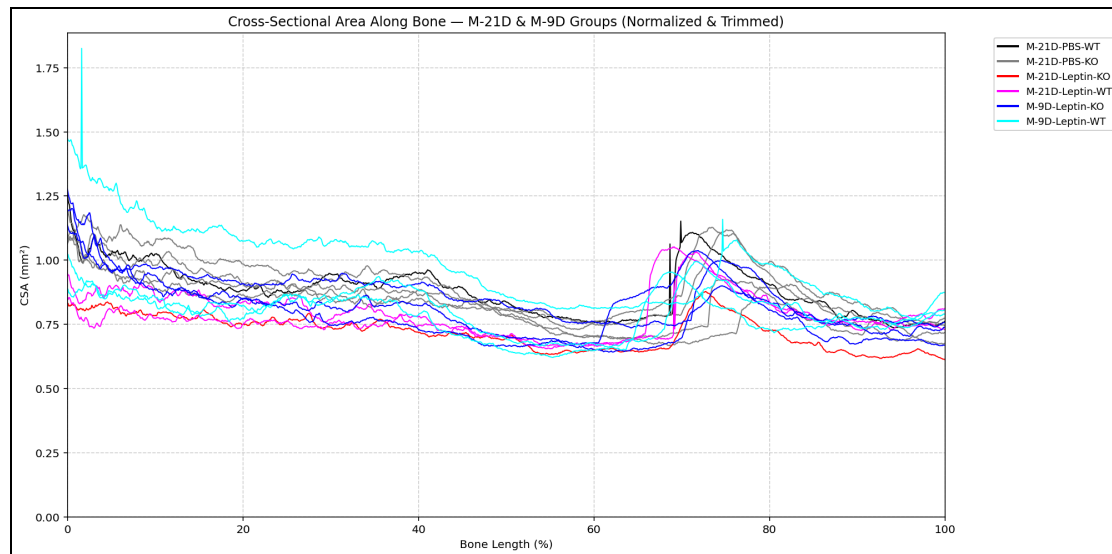


Figure 11: Cross sectional area for 21 day and 9 day treated male bones across the bone length (normalized).

Cross Sectional Area:

According to Figure 11, leptin treatment does not induce a uniform increase in cross-sectional area (CSA) along the tibial length. CSA profiles for leptin-treated and PBS-treated groups largely overlap across most regions of the bone, with no consistent global elevation observed in treated groups. While localized deviations are present in specific regions, these differences are not consistent across treatment durations or genotypes.

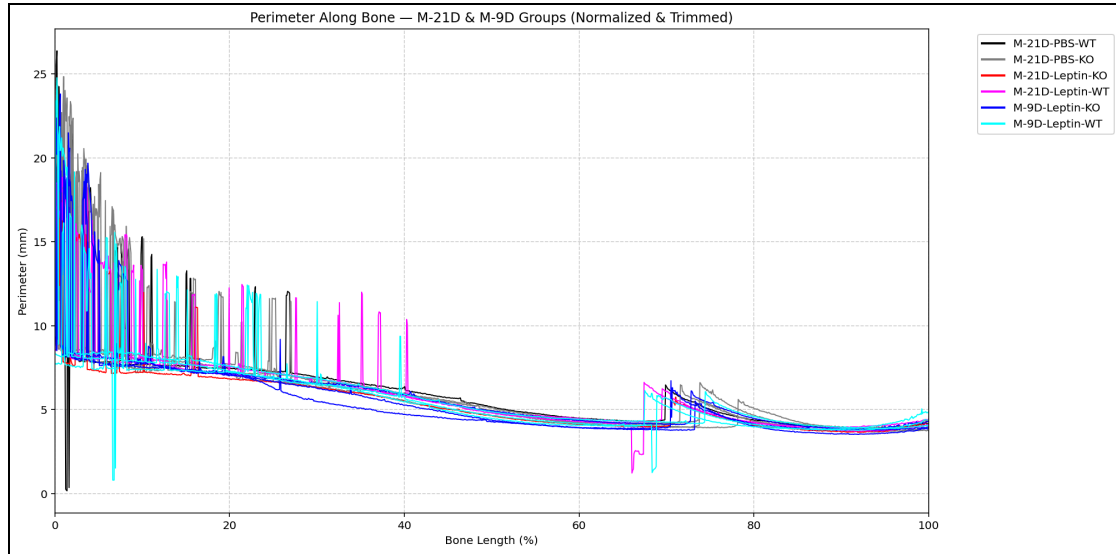


Figure 12: Perimeter for 21 day and 9 day treated male bones across the bone length (normalized).

Perimeter:

Figure 12 displays perimeter measurements along the normalized bone length for all treatment groups. Despite the presence of localized spikes, particularly near the proximal region of the tibia, perimeter values largely overlap among leptin-treated and PBS-treated groups across most of the bone. No consistent global differences are observed between treatment durations or genotypes, indicating that leptin treatment did not produce measurable changes in the external cortical boundary.

Conclusion and Future Steps:

Taurine Project:

In the taurine study, mechanical loading produced clear increases in cortical thickness compared to non-loaded controls, consistent with established mechanoadaptive responses in the tibia. Mechanical loading was also associated with increases in bone volume, indicating that loading promoted overall bone accrual in addition to geometric adaptation.

In contrast, taurine treatment did not significantly alter cortical thickness or bone volume either in the absence of loading or when combined with mechanical loading. Comparisons between taurine-treated and vehicle-treated loaded groups revealed no additional effect of taurine beyond the structural changes induced by loading alone.

Together, these results indicate that mechanical loading was the primary driver of both cortical thickness and bone volume changes, while taurine supplementation did not measurably influence these outcomes under the conditions tested.

Future work in the taurine project should assess whether treatment duration or dosage alters its interaction with mechanical loading. Increasing sample size and incorporating molecular analyses may help clarify whether taurine affects bone metabolism without producing measurable changes in cortical structure.

Leptin Project:

Across both male and female mice, leptin treatment resulted in a reduction in cortical thickness along the length of the tibia when compared to PBS-treated controls. This effect was observed in both Wild-Type (WT) and Knockout (KO) groups, suggesting that leptin supplementation is associated with decreased cortical thickness independent of genotype.

Additionally, comparison between treatment durations revealed that 21-day leptin treatment produced a greater reduction in cortical thickness than 9-day treatment, indicating that leptin's effects on cortical bone are duration-dependent.

Due to the limited number of mice per group, definitive conclusions regarding genotype-dependent differences between WT and KO mice could not be drawn and should be interpreted cautiously. Although Wild-Type and Knockout mice exhibited similar trends across cortical thickness, CSA, I_{max}, and I_{min} measurements, the limited sample size restricts the ability to draw definitive genotype-dependent conclusions. The largely overlapping structural profiles suggest that leptin's effects on cortical bone geometry may not be strongly mediated by adipocyte-driven energy mobilization under the conditions tested. However, subtle

genotype-specific effects may be masked by variability and warrant further investigation with expanded cohorts.

Analysis of cross-sectional area (CSA) along the tibial length did not reveal a consistent global increase associated with leptin treatment, though localized, region-specific differences were observed. Similarly, perimeter measurements along the tibial length showed substantial overlap between leptin-treated and PBS control groups across both durations and genotypes, with no consistent global changes observed. These findings suggest that leptin treatment did not substantially alter the external cortical boundary of the tibia.

Similarly, evaluation of bending resistance parameters (I_{max} and I_{min}) showed modest, region-dependent variations without a clear, uniform enhancement in structural strength. Together with the observed reductions in cortical thickness, these results indicate that leptin treatment alters cortical bone geometry in a space dependent manner, rather than promoting overall structural reinforcement. The absence of consistent changes in perimeter further supports this conclusion, indicating that leptin-associated structural changes occur without significant expansion.

In the future, increasing sample size would allow more robust assessment of WT versus KO differences in response to leptin treatment. Future work related to the leptin project should also try and directly quantify bone marrow adipose tissue to better understand the metabolic mechanisms underlying leptin-induced changes in cortical bone.

References:

Works Cited

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Additionally, thank you Dr. Erica Scheller for letting me work in your lab, and creating such a positive and inclusive environment.

Through this independent study, I gained hands-on experience in quantitative bone imaging, data normalization, and structural analysis. I developed proficiency in microCT analysis, parameter selection, and interpretation of biomechanical metrics, as well communicating the data I extracted through figure generation and meetings with my lab PI, Dr. Erica Scheller.

I created all of the python plots (figures 1 + 3-12) included in the report.